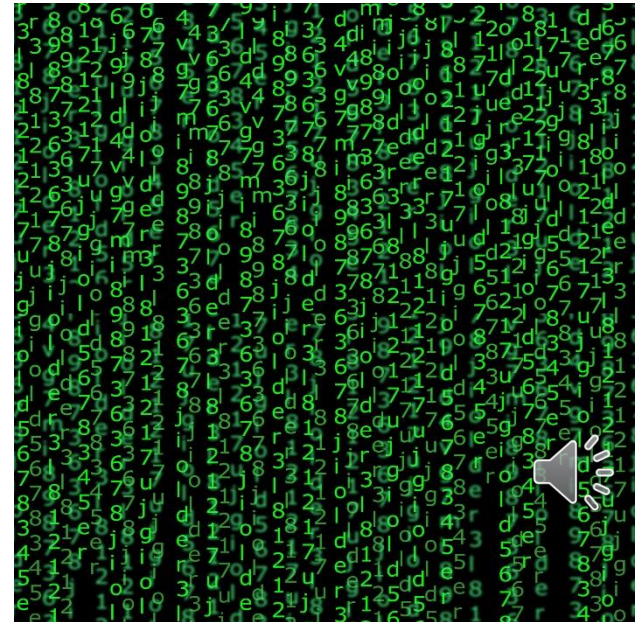


Week9: Matrix Algebra



INF1004 Linear Algebra (Part II) Week9: Matrices



Learning Outcomes

After studying this chapter, you should

1. Be able to conduct the basic operations of matrices
2. Be able to expand 2×2 determinant
3. Solve pairs of simultaneous linear equations in two variables using 2×2 determinants
4. Be able to expand 3×3 determinant.
5. Find adjoint, cofactor, inverse of a matrix.
6. Express system of linear equations in matrix form.



Matrices - The 'Things' of Linear Algebra

Generally, a matrix is just a 'group' (or array) of elements. A matrix with **m rows** and **n columns** is read as an **m** by **n** matrix (or $m \times n$ matrix).

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

where the a_{ij} are the elements.

A matrix that is $m \times 1$ is also called a vector.

Definition of A System of Linear Equations

Generally, a system of linear equations has the form:

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\&\vdots \\a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m\end{aligned}$$

where there are **m equations** with **n unknown variables, x_1 to x_n** . The a_{ij} are constant coefficients and the b_i are constants of the system.

System of Linear Equations (SLE)

Examples:

$$\begin{aligned}2x + 3y &= 1 \\ x - 7y &= -14\end{aligned}$$

(A system with 2 variables: x & y)

$$\begin{aligned}2x + y - z &= 1 \\ 2x - 5y - z &= 7 \\ x + y + z &= 1\end{aligned}$$

(A system with 3 variables: x , y & z)

$$\begin{aligned}w_{11}x_1 + w_{12}x_2 + w_{13}x_3 + w_{14}x_4 &= b_1 \\ w_{21}x_1 + w_{22}x_2 + w_{23}x_3 + w_{24}x_4 &= b_2 \\ w_{31}x_1 + w_{32}x_2 + w_{33}x_3 + w_{34}x_4 &= b_3\end{aligned}$$

(A system with 4 variables: x_1 , x_2 , x_3 & x_4)

System of Linear Equations (SLE)

Matrices and determinants are mainly used for the solution of linear simultaneous equations.

Real world problems can give rise to a SLE, where its solution possibly represents the desired condition to achieve a 'best' outcome.

You should be quite familiar with the method of solving a pair of simultaneous equations by elimination. For example,

$$2x + 3y = -2 \quad (1)$$

$$3x + 4y = -6 \quad (2)$$

$$[ans: x = -10, y = 6]$$

In this chapter, we will use matrix algebra to tackle the same problem.

System of Linear Equations (SLE)

A matrix is a rectangular array of numbers, symbols, or expressions, arranged in rows and columns.

For example, a table of numbers

5	1	0
5	1	0
4	1	1

Can be represented by a “matrix”

$$\begin{bmatrix} 5 & 1 & 0 \\ 5 & 1 & 0 \\ 4 & 1 & 1 \end{bmatrix}$$

Excel, tables and many data or database are in matrix form

System of Linear Equations (SLE)

Example 1: $2x + 3y = -2$ (1)

$3x + 4y = -6$ (2)

The coefficients of the variables for linear simultaneous equations may be shown in matrix form as below.

$$\begin{pmatrix} 2 & 3 \\ 3 & 4 \end{pmatrix}$$

Similarly, for

$1.3p - 2.0q + r = 7$ (1)

$3.7p + 4.8q - 7r = 3$ (2)

$4.1p + 3.8q + 12r = -6$ (3)

$$\begin{pmatrix} 1.3 & -2.0 & 1 \\ 3.7 & 4.8 & -7 \\ 4.1 & 3.8 & 12 \end{pmatrix}$$

System of Linear Equations (SLE)

- Matrices (the plural of matrix) may be used to display information and to solve systems of linear equations.
- The numbers (or symbols, expressions) in the rows and columns of a matrix are called the **elements** of the matrix.
- The dimension of a matrix is determined by the number of its rows and the number of its columns

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{pmatrix}$$

2/25/2026 has m rows and n columns. It is an **m × n** matrix.

System of Linear Equations (SLE)

- If $m = n$, then the array is square, and **A** is called a **square matrix** of order n .
- If the matrix has one column or one row, then it is called a **column vector** or a **row vector**, respectively.

Row vector: $\mathbf{a} = [a_1, a_2, \dots, a_n]$

Column vector: $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$

Special matrices

- The **unit matrix or Identity**, **I**, is a square matrix whose only non-zero elements are on the diagonal and are equal to one.

$$\mathbf{I} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{I} = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 & 0 \\ 0 & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & 0 \\ 0 & 0 & 0 & \dots & 0 & 1 \end{pmatrix}$$

- All elements of the **zero matrix**, **0**, are equal to zero.

$$\mathbf{0} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{0} = \begin{pmatrix} 0 & 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & 0 & \dots & 0 & 0 \end{pmatrix}$$

- A **diagonal matrix**, **D** only has non-zero elements on the main diagonal. These non-zero elements can have any value.

$$\mathbf{D} = \begin{pmatrix} d_{11} & 0 & 0 \\ 0 & d_{22} & 0 \\ 0 & 0 & d_{33} \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} d_{11} & 0 & \dots & 0 & 0 \\ 0 & d_{22} & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & d_{n-1,n-1} & 0 \\ 0 & 0 & \dots & 0 & d_{nn} \end{pmatrix}$$

Basic operations of matrices

(a) Equality

Two matrices are equal if and only if they have the same size and all their elements are the same.

$$\mathbf{A} = \mathbf{B}$$

$$a_{ij} = b_{ij} \quad \text{for } i = 1, \dots, m; j = 1, \dots, n$$

Basic operations of matrices

(b) Addition

Two matrices can only be added if they have the same size. Corresponding elements in two matrices are added to form a single matrix.

Example 2:

Perform the addition

$$\begin{pmatrix} 3 & 1 & -4 \\ 4 & 3 & 1 \\ 1 & 4 & -3 \end{pmatrix} + \begin{pmatrix} 2 & 7 & -5 \\ -2 & 1 & 0 \\ 6 & 3 & 4 \end{pmatrix} = \begin{pmatrix} 5 & 8 & -9 \\ 2 & 4 & 1 \\ 7 & 7 & 1 \end{pmatrix}$$

Basic operations of matrices

(c) Subtraction

If A is a matrix and B is another matrix, then $(A-B)$ is a single matrix formed by subtracting the elements of B from the corresponding elements of A.

Two matrices can only be added if they have the same size

Example 3:

$$\begin{pmatrix} 3 & 1 & -4 \\ 4 & 3 & 1 \\ 1 & 4 & -3 \end{pmatrix} - \begin{pmatrix} 2 & 7 & -5 \\ -2 & 1 & 0 \\ 6 & 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 & -6 & 1 \\ 6 & 2 & 1 \\ -5 & 1 & -7 \end{pmatrix}$$

Basic operations of matrices

(d) Multiplication by a scalar

Multiply each element by the scalar number.

Example 4:

$$2 \begin{pmatrix} 3 & 1 & -4 \\ 4 & 3 & 1 \\ 1 & 4 & -3 \end{pmatrix} = \begin{pmatrix} 6 & 2 & -8 \\ 8 & 6 & 2 \\ 2 & 8 & -6 \end{pmatrix}$$

Basic Operation of Matrices

(e) Properties of the transpose

Let $\mathbf{A}$ be an $m \times n$ matrix, with elements a_{ij} . Then the transpose of $\mathbf{A}$, written $\mathbf{A}^T$ is the $n \times m$ matrix with elements a_{ji} .

Example 5:

$$\mathbf{A} = \begin{pmatrix} 2 & 3 \\ 1 & 2 \\ 4 & 5 \end{pmatrix} \quad 3 \times 2 \text{ matrix}$$

$$\mathbf{A}^T = \begin{pmatrix} 2 & 1 & 4 \\ 3 & 2 & 5 \end{pmatrix} \quad 2 \times 3 \text{ matrix}$$

Some facts about transpose:

- $(\mathbf{A}^T)^T = \mathbf{A}$
- $(\mathbf{A} + \mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T$
- $(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$
- A matrix is called symmetric if $\mathbf{A}^T = \mathbf{A}$
- A matrix is called skew-symmetric if $\mathbf{A}^T = -\mathbf{A}$

Basic operations of matrices

(f) Basic rules of addition

Commutative law

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$$

Associative law

$$(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$$

Distributive law

$$\lambda(\mathbf{A} + \mathbf{B}) = \lambda\mathbf{A} + \lambda\mathbf{B}$$

Matrix Operations

Example 6: Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ 1 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix}, \mathbf{C} = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

Find, where possible,

- (a) $\mathbf{A} + \mathbf{B}$,
- (b) $\mathbf{A} + \mathbf{C}$,
- (c) $\mathbf{C} - \mathbf{A}$,
- (d) $3\mathbf{A}$,
- (e) $4\mathbf{B}$,
- (f) $\mathbf{C} + \mathbf{B}$,
- (g) $3\mathbf{A} + 2\mathbf{C}$,
- (h) $\mathbf{A}^T + \mathbf{A}$
- (i) $\mathbf{A} + \mathbf{C}^T + \mathbf{B}^T$.

Matrix Operations

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ 1 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix}, \mathbf{C} = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

(a) $\mathbf{A} + \mathbf{B}$ is not possible.

(b)

$$\mathbf{A} + \mathbf{C} = \begin{pmatrix} 1 & 3 & 2 \\ 1 & 1 & 3 \\ 2 & 1 & 1 \end{pmatrix}$$

(c)

$$\mathbf{C} - \mathbf{A} = \begin{pmatrix} -1 & -1 & 0 \\ -1 & -1 & -1 \\ 0 & -1 & -1 \end{pmatrix}$$

(d)

$$3\mathbf{A} = \begin{pmatrix} 3 & 6 & 3 \\ 3 & 3 & 6 \\ 3 & 3 & 3 \end{pmatrix}$$

(e)

$$4\mathbf{B} = \begin{pmatrix} 8 & 4 \\ 4 & 0 \\ 4 & 4 \end{pmatrix}$$

Matrix Operations

Please work on the rest of the question.

Matrix Operations

(f) $\mathbf{C} + \mathbf{B}$ is not possible.

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ 1 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix}, \mathbf{C} = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

(g)

$$3\mathbf{A} + 2\mathbf{C} = \begin{pmatrix} 3 & 6 & 3 \\ 3 & 3 & 6 \\ 3 & 3 & 3 \end{pmatrix} + \begin{pmatrix} 0 & 2 & 2 \\ 0 & 0 & 2 \\ 2 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 3 & 8 & 5 \\ 3 & 3 & 8 \\ 5 & 3 & 3 \end{pmatrix}$$

(h)

$$\mathbf{A}^T + \mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 2 & 1 \\ 1 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 3 & 2 \\ 3 & 2 & 3 \\ 2 & 3 & 2 \end{pmatrix}$$

(i) $\mathbf{A} + \mathbf{C}^T + \mathbf{B}^T$ is not possible.

Matrix Multiplication

If **A** is an $m \times p$ matrix with elements a_{ij} and **B** a $p \times n$ matrix with elements b_{ij} then we define the product **C** = **AB** as the $m \times n$ matrix with components

$$C_{ij} = \sum_{k=1}^p a_{ik}b_{kj} = a_{i1}b_{1j} + \cdots + a_{ip}b_{pj}, \quad i = 1, \dots, m, \quad j = 1, \dots, n$$

Matrix Multiplication

Example 7:

$$\begin{array}{c}
 \boxed{\begin{bmatrix} 2 & 3 & 1 \\ -1 & 2 & 4 \end{bmatrix}} \quad \boxed{\begin{bmatrix} 5 & 3 \\ 4 & 0 \\ 3 & 1 \end{bmatrix}} = \begin{bmatrix} 2 \times 5 + 3 \times 4 + 1 \times 3 & 2 \times 3 + 3 \times 0 + 1 \times 1 \\ -1 \times 5 + 2 \times 4 + 4 \times 3 & -1 \times 3 + 2 \times 0 + 4 \times 1 \end{bmatrix} \\
 (2 \times 3) \quad (3 \times 2) \quad = \begin{bmatrix} 25 & 7 \\ 15 & 1 \end{bmatrix} \quad (2 \times 2)
 \end{array}$$

Matrix Multiplication

Given

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \\ 1 & 3 \end{pmatrix}, \mathbf{C} = \begin{pmatrix} 1 & -2 \\ -1 & 2 \\ -2 & 4 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}, \mathbf{c} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

Find, (a) $\mathbf{AB}$, (b) $\mathbf{BA}$, (c) $\mathbf{Bb}$, (d) $\mathbf{A}^T\mathbf{b}$, (e) $\mathbf{c}^T(\mathbf{A}^T\mathbf{b})$ and (f) $\mathbf{AC}$.

Matrix Multiplication

(a) $\mathbf{AB}$

$$= \begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \\ 1 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} (1)(2) + (1)(0) + (0)(1) & (1)(0) + (1)(1) + (0)(3) \\ (2)(2) + (0)(0) + (1)(1) & (2)(0) + (0)(1) + (1)(3) \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$$

(b) $\mathbf{BA}$

$$= \begin{pmatrix} 2 & 0 \\ 0 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 0 & 1 \\ 7 & 1 & 3 \end{pmatrix}$$

(c) $\mathbf{Bb}$

$$= \begin{pmatrix} 2 & 0 \\ 0 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} -1 \\ 2 \end{pmatrix} = \begin{pmatrix} -2 \\ 2 \\ 5 \end{pmatrix}$$

Matrix Multiplication

$$(d) \quad \mathbf{A}^T \mathbf{b} = \begin{pmatrix} 1 & 2 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$$

$$(e) \quad \mathbf{c}^T (\mathbf{A}^T \mathbf{b}) = (1 \quad 1 \quad -1) \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = (3 - 1 - 2) \\ = 0$$

$$(f) \quad \mathbf{A} \mathbf{C} = \begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ -1 & 2 \\ -2 & 4 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \\ = 0$$

Properties of Matrix Multiplication

(a) Commutative law

$$AB \neq BA$$

(b) Associative law

$$A(BC) = (AB)C$$

where **A** is $m \times p$, **B** is $p \times q$ and **C** is $q \times n$.

(c) Distributive law over addition

$$(A + B)C = AC + BC \quad \text{and} \quad A(B + C) = AB + AC$$

(d) Multiplication by unit matrices

$$IA = A = AI$$

(e) Transpose of a product

$$(AB)^T = B^T A^T$$

Determinants

The idea of a determinant is crucial to the solution of linear equations.

(a) 2 x 2 determinant

Given a square matrices

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

The determinant of $\mathbf{A}$, denoted by $\det \mathbf{A}$ or $|\mathbf{A}|$, is given by

$$|\mathbf{A}| = ad - bc$$

Determinants

Example 9:

Determine the value of $\begin{vmatrix} 3 & -2 \\ 7 & 4 \end{vmatrix}$.

Solution

$$\begin{aligned} & \begin{vmatrix} 3 & -2 \\ 7 & 4 \end{vmatrix} \\ &= (3 \times 4) - (-2 \times 7) \\ &= 12 - (-14) = 26 \end{aligned}$$

Determinants

Example 10:

Evaluate $\begin{vmatrix} 1 & 2 \\ -3 & -4 \end{vmatrix}$

Solution

$$\begin{vmatrix} 1 & 2 \\ -3 & -4 \end{vmatrix} = 1(-4) - (-3)2 \\ = -4 + 6 = 2$$

Determinants

(b) 3 x 3 determinant

(i) The **minor** of an element of a 3 by 3 matrix is the value of the 2 by 2 determinant obtained by covering up the row and column containing that element.

For example,

$$\det(\mathbf{A}) = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

(Note: In the original image, the element a_{22} is highlighted in red, and the entire 3x3 matrix is overlaid with a light blue cross shape representing the removal of row 2 and column 2 to find the minor.)

$$M_{22} = \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}$$

Determinants

For instance, the **minor of element 4** in the matrix

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \text{ is obtained by covering the row } (4 \ 5 \ 6) \text{ and the column } \begin{pmatrix} 1 \\ 4 \\ 7 \end{pmatrix},$$

leaving the 2 by 2 determinant $\begin{vmatrix} 2 & 3 \\ 8 & 9 \end{vmatrix}$.

The minor of element 4 is $(2 \times 9) - (3 \times 8) = 18 - 24 = -6$

Determinants

(ii) The **sign of a minor** depends on its position within the matrix, the sign

pattern being $\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$.

Thus, the signed-minor of element 4 is $- \begin{vmatrix} 2 & 3 \\ 8 & 9 \end{vmatrix} = -(-6) = 6$

A minor multiplied by the appropriate sign is called the **cofactor** of the element.

In general, the cofactor $A_{ij} = (-1)^{(i+j)} M_{ij}$

$$A_{22} = (-1)^{(2+2)} M_{22} = +M_{22}$$

$$A_{23} = (-1)^{(2+3)} M_{23} = -M_{23}$$

Determinants

(iii) The value of a 3 by 3 determinant is the sum of the products of the elements and their cofactors of any row or any column of the corresponding 3 by 3 matrix.

There are six different ways of evaluating a 3 x 3 determinant and all give the same value.

Determinants

Example 11:

Determine the value of $\begin{vmatrix} 3 & 4 & -1 \\ 2 & 0 & 7 \\ 1 & -3 & -2 \end{vmatrix}$.

Determinants

Choose the 1st row:

$$\begin{vmatrix} 3 & 4 & -1 \\ 2 & 0 & 7 \\ 1 & -3 & -2 \end{vmatrix} = 3 \begin{vmatrix} 0 & 7 \\ -3 & -2 \end{vmatrix} - 4 \begin{vmatrix} 2 & 7 \\ 1 & -2 \end{vmatrix} - 1 \begin{vmatrix} 2 & 0 \\ 1 & -3 \end{vmatrix}$$

$$= 3(21) - 4(-11) - 1(-6)$$

$$= 63 + 44 + 6$$

$$= 113$$

Or choose the 2nd row:

$$\begin{vmatrix} 3 & 4 & -1 \\ 2 & 0 & 7 \\ 1 & -3 & -2 \end{vmatrix} = -2 \begin{vmatrix} 4 & -1 \\ -3 & -2 \end{vmatrix} + 0 \begin{vmatrix} 3 & -1 \\ 1 & -2 \end{vmatrix} - 7 \begin{vmatrix} 3 & 4 \\ 1 & -3 \end{vmatrix}$$

$$= -2(-11) + 0 - 7(-13)$$

$$= 22 + 91$$

$$= 113$$

Determinants

Your practice:

1. Choose the 3rd row and calculate
2. Choose the 2nd column and calculate

Determinants

Example 12:

Determine the value of

$$\begin{vmatrix} 2 & 1 & 3 \\ -1 & 1 & 1 \\ 0 & 4 & 5 \end{vmatrix}$$

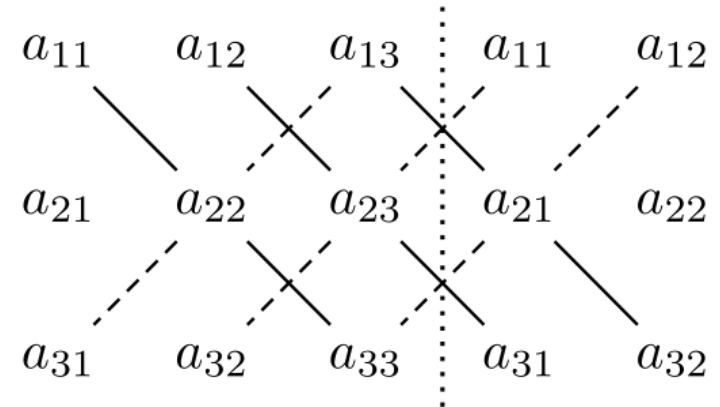
Solution

$$\begin{aligned} \begin{vmatrix} 2 & 1 & 3 \\ -1 & 1 & 1 \\ 0 & 4 & 5 \end{vmatrix} &= 0 \begin{vmatrix} 1 & 3 \\ 1 & 1 \end{vmatrix} - 4 \begin{vmatrix} 2 & 3 \\ -1 & 1 \end{vmatrix} + 5 \begin{vmatrix} 2 & 1 \\ -1 & 1 \end{vmatrix} \\ &= -4 [2(1) - (-1)(3)] + 5 [2(1) - (-1)(1)] \\ &= -4(2 + 3) + 5(2 + 1) \\ &= -20 + 15 = -5 \end{aligned}$$

Rule of Sarrus

Sarrus's rule: The determinant of the three columns on the left is the sum of the products along the solid diagonals minus the sum of the products along the dashed diagonals

Sarrus's rule

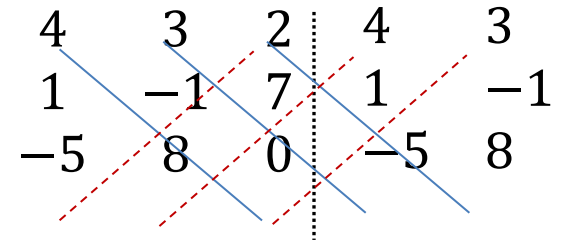


$$\det(M) = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}.$$

Rule of Sarrus

RULE OF SARRUS



$$\begin{vmatrix} 4 & 3 & 2 \\ 1 & -1 & 7 \\ -5 & 8 & 0 \end{vmatrix} = [4 \times (-1) \times 0 + 3 \times 7 \times (-5) + 2 \times 1 \times 8] \\
 - [(-5) \times (-1) \times 2 + 8 \times 7 \times 4 + 0 \times 1 \times 3] \\
 = -323$$

LUR: Revisit vector (cross) product

Better way to remember:

$$\begin{aligned}\mathbf{a} \times \mathbf{b} &= \begin{vmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = i \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - j \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + k \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \\ &= (a_2 b_3 - a_3 b_2)i + (a_3 b_1 - a_1 b_3)j + (a_1 b_2 - a_2 b_1)k\end{aligned}$$

We now can explain more on the matrices

Answer to Leading Question

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = i \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - j \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + k \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}$$

$$\begin{aligned} (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} &= \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} c_1 - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} c_2 + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} c_3 \\ &= \begin{vmatrix} c_1 & c_2 & c_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}. \end{aligned}$$

Adjoint Matrix

It is defined as the transpose of the matrix of cofactors.

$$\text{adj}(\mathbf{A}) = \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ A_{n1} & A_{n2} & \cdots & A_{nn} \end{bmatrix}^T$$

Steps

1. Compute the Minors
2. Obtain Cofactors (Minors with associated signs)
3. Determine the transpose of the Cofactor Matrix

Adjoint Matrix

(a) For a 2 x 2 matrix

$$\mathbf{A} = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$$

$$\text{adj}\mathbf{A} = \begin{pmatrix} s & -q \\ -r & p \end{pmatrix}$$

1. Minors: s, r, q, p

$$\begin{bmatrix} s & r \\ q & p \end{bmatrix}$$

2. Cofactors

$$\begin{bmatrix} +s & -r \\ -q & +p \end{bmatrix}$$

3. Transpose

$$\begin{bmatrix} +s & -q \\ -r & +p \end{bmatrix}$$

The adjoint matrix may be obtained by:

- (i) Interchanging the positions of p and s ;
- (ii) Changing the signs of q and r .

Example 13

Given

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$$

$$\text{adj}\mathbf{A} = \begin{pmatrix} 3 & -2 \\ -2 & 1 \end{pmatrix}$$

Adjoint Matrix (Summary)

(b) For a 3 x 3 matrix

The adjoint matrix may be obtained by:

- (i) Forming a matrix **B** of the cofactors of **A**, and
- (ii) Transposing matrix **B** to give **B^T**.

Adjoint Matrix

Example 14:

Given $\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 0 & 1 \\ 3 & 1 & 1 \end{pmatrix}$, find the *adj* $\mathbf{A}$.

Solution

adj $\mathbf{A}$

$$= \begin{pmatrix} \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} & -\begin{vmatrix} 2 & 1 \\ 3 & 1 \end{vmatrix} & \begin{vmatrix} 2 & 0 \\ 3 & 1 \end{vmatrix} \\ -\begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} & \begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} & -\begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix} \\ \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} & -\begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 2 & 0 \end{vmatrix} \end{pmatrix}^T = \begin{pmatrix} -1 & 1 & 2 \\ 1 & -5 & 2 \\ 1 & 3 & -2 \end{pmatrix}^T = \begin{pmatrix} -1 & 1 & 1 \\ 1 & -5 & 3 \\ 2 & 2 & -2 \end{pmatrix}$$

The Inverse Matrix

Given a square matrix **A**, if we can construct a matrix **B** such that:

$$\mathbf{BA} = \mathbf{AB} = \mathbf{I}$$

Then **B** is called the **inverse matrix of A**, and write it as **A⁻¹**.

$$\mathbf{A}^{-1} = \frac{\text{adj}\mathbf{A}}{|\mathbf{A}|}$$

- *For the inverse to exist, matrix A must be a **square matrix***
- *The inverse matrix is also a square matrix*
- **The inverse of a square matrix may not always exist**

The Inverse Matrix

Example 16:

Determine the inverse of the matrix

$$\mathbf{A} = \begin{pmatrix} 3 & 4 & -1 \\ 2 & 0 & 7 \\ 1 & -3 & -2 \end{pmatrix}$$

The Inverse Matrix

$$|\mathbf{A}| = 3 \begin{vmatrix} 0 & 7 \\ -3 & -2 \end{vmatrix} - 4 \begin{vmatrix} 2 & 7 \\ 1 & -2 \end{vmatrix} - 1 \begin{vmatrix} 2 & 0 \\ 1 & -3 \end{vmatrix} = 3(21) - 4(-11) - 1(-6)$$

$$= 63 + 44 + 6 = 113$$

$$\mathbf{A} = \begin{pmatrix} 3 & 4 & -1 \\ 2 & 0 & 7 \\ 1 & -3 & -2 \end{pmatrix}$$

$$\text{adj}\mathbf{A} = \begin{pmatrix} \begin{vmatrix} 0 & 7 \\ -3 & -2 \end{vmatrix} & -\begin{vmatrix} 2 & 7 \\ 1 & -2 \end{vmatrix} & \begin{vmatrix} 2 & 0 \\ 1 & -3 \end{vmatrix} \\ -\begin{vmatrix} 4 & -1 \\ -3 & -2 \end{vmatrix} & \begin{vmatrix} 3 & -1 \\ 1 & -2 \end{vmatrix} & -\begin{vmatrix} 3 & 4 \\ 1 & -3 \end{vmatrix} \\ \begin{vmatrix} 4 & -1 \\ 0 & 7 \end{vmatrix} & -\begin{vmatrix} 3 & -1 \\ 2 & 7 \end{vmatrix} & \begin{vmatrix} 3 & 4 \\ 2 & 0 \end{vmatrix} \end{pmatrix}^T = \begin{pmatrix} 21 & 11 & -6 \\ 11 & -5 & 13 \\ 28 & -23 & -8 \end{pmatrix}^T = \begin{pmatrix} 21 & 11 & 28 \\ 11 & -5 & -23 \\ -6 & 13 & -8 \end{pmatrix}$$

$$\mathbf{A}^{-1} = \frac{\text{adj}\mathbf{A}}{|\mathbf{A}|} = \frac{1}{113} \begin{pmatrix} 21 & 11 & 28 \\ 11 & -5 & -23 \\ -6 & 13 & -8 \end{pmatrix}$$

The Inverse Matrix

Example 15:

Given

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \quad \text{and} \quad \mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

Evaluate $(\mathbf{AB})^{-1}$, $\mathbf{A}^{-1}\mathbf{B}^{-1}$, $\mathbf{B}^{-1}\mathbf{A}^{-1}$ and show that $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$.

The Inverse Matrix

$$\mathbf{AB} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 1 & 3 \end{pmatrix}$$

$$(\mathbf{AB})^{-1} = \frac{1}{3} \begin{pmatrix} 3 & -3 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -\frac{1}{3} & \frac{2}{3} \end{pmatrix}$$

$$\mathbf{A}^{-1} = \frac{1}{-3} \begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix}$$

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

$$\mathbf{B}^{-1} = \frac{1}{-1} \begin{pmatrix} 1 & -1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

$$\mathbf{A}^{-1}\mathbf{B}^{-1} = \frac{1}{-3} \begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} = \frac{1}{-3} \begin{pmatrix} -3 & 1 \\ 3 & -2 \end{pmatrix}$$

$$\mathbf{B}^{-1}\mathbf{A}^{-1} = \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} \frac{1}{-3} \begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix} = \frac{1}{-3} \begin{pmatrix} -3 & 3 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -\frac{1}{3} & \frac{2}{3} \end{pmatrix} = (\mathbf{AB})^{-1}$$

The Inverse Matrix

Let A be an $n \times n$ matrix. If there exists an $n \times n$ matrix A^{-1} (read: “A inverse”) such that

$$AA^{-1} = I_n \text{ and } A^{-1}A = I_n,$$

is the multiplicative inverse of A .

If a square matrix has a multiplicative inverse, it is said to be **invertible** or **nonsingular**. If a square matrix has no multiplicative inverse, it is called **singular**.